

The Mathematical Model: Differential Equations

$$\frac{dC_i}{dt} = \alpha_i C_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j C_j - k_i C_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

$$E_i = \lambda_i C_i$$

$$i \in \{1, 2, \dots, N_{Regions}\}$$

$$P_i(t), N_i(t) > 0$$

$$E_i, C_i > 0$$

$$\alpha_i, k_i, \lambda_i > 0$$

Finding $P_i(t)$ and $N_i(t)$: Task (write)

We will find $N_i(t)$ via linear regression using the data points on number of citizens in region i from 2021 to 2026 and $P_i(t)$ via linear interpolation using the police population data points from 2021 to 2026.

Finding α_i, k_i, α_j : Task (write)

We will find α_i, k_i, α_j via solving integral equations using linear least squares.

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) \quad (\text{reasoning given in the sections below})$$

below)

$E_i(m_1)$ = data of crimes committed in the initial month in region i

Note: If two regions don't border then $\alpha_j = 0$. In the code, I am not sure but, $t=0$ first month $t=23$ 24th month...

! It is crucial that timelines match

! We are modeling the expected number of criminals C_i . And expected number of crimes committed in a month E_i . When we know its data, E_i equals the data as shown in the explanation Explanation for $E_i = \lambda_i C_i$ section.

m_1 month 1

$E_i(m_1)$ number of crimes committed in month 1 in region i

$$\int_{m_1}^{m_2} \frac{dE_i}{dt} dt = E_i(m_2) - E_i(m_1)$$

We solve for all months we have

$$\int_{m_{c-1}}^{m_c} \frac{dE_i}{dt} dt = E_i(m_c) - E_i(m_{c-1})$$

By linearity of integration:

$$\begin{aligned} \int_{m_a}^{m_{a+1}} \frac{dE_i}{dt} dt &= \int_{m_a}^{m_{a+1}} E_i(\alpha_i - k_i (\frac{P_i(t)}{N_i(t) + P_i(t)})) dt + \int_{m_a}^{m_{a+1}} \sum_{j=1, j \neq i}^{N_R} \alpha_j E_j dt \\ &= \alpha_i \int_{m_a}^{m_{a+1}} E_i dt - k_i \int_{m_a}^{m_{a+1}} E_i (\frac{P_i(t)}{N_i(t) + P_i(t)}) dt + \sum_{j=1, j \neq i}^{N_R} \alpha_j \int_{m_a}^{m_{a+1}} E_j dt \end{aligned}$$

If we have K amount of months (K amount of monthly data for E_i). Then:

Simultaneous Integral Equations:

$$\int_{m_1}^{m_2} \frac{dE_i}{dt} dt = E_i(m_2) - E_i(m_1)$$

$$\int_{m_2}^{m_3} \frac{dE_i}{dt} dt = E_i(m_3) - E_i(m_2)$$

...

$$\int_{m_{K-1}}^{m_K} \frac{dE_i}{dt} dt = E_i(m_K) - E_i(m_{K-1})$$

Which can be expressed using matrices and vectors:

$$\begin{bmatrix} E_i(m_2)-E_i(m_1) \\ E_i(m_3)-E_i(m_2) \\ \dots \\ E_i(m_K)-E_i(m_{K-1}) \end{bmatrix} = \begin{bmatrix} \int_{m_1}^{m_2} E_i dt & \int_{m_1}^{m_2} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_1}^{m_2} E_j dt & \dots & \int_{m_1}^{m_2} E_j dt \\ \int_{m_2}^{m_3} E_i dt & \int_{m_2}^{m_3} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_2}^{m_3} E_j dt & \dots & \int_{m_2}^{m_3} E_j dt \\ \dots & \dots & \dots & \dots & \dots \\ \int_{m_{K-1}}^{m_K} E_i dt & \int_{m_{K-1}}^{m_K} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_{K-1}}^{m_K} E_j dt & \dots & \int_{m_{K-1}}^{m_K} E_j dt \end{bmatrix} \begin{bmatrix} \alpha_i \\ k_i \\ \alpha_j \dots \\ \alpha_{N_R} \end{bmatrix}$$

Which is a linear least squares problem:

$$\overline{b}_i = A_i \overline{x}_i$$

Which has the solution formula: (where superscript -1 means matrix inverse and superscript T means transpose of the matrix)

$$\overline{x}_i = (A_i^T A_i)^{-1} A_i^T \overline{b}_i$$

Where:

$$\overline{b}_i = \begin{bmatrix} E_i(m_2)-E_i(m_1) \\ E_i(m_3)-E_i(m_2) \\ \dots \\ E_i(m_K)-E_i(m_{K-1}) \end{bmatrix}$$

$(K - 1) \times 1$ vector

$$A_i = \begin{bmatrix} \int_{m_1}^{m_2} E_i dt & \int_{m_1}^{m_2} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_1}^{m_2} E_j dt & \dots & \int_{m_1}^{m_2} E_j dt \\ \int_{m_2}^{m_3} E_i dt & \int_{m_2}^{m_3} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_2}^{m_3} E_j dt & \dots & \int_{m_2}^{m_3} E_j dt \\ \dots & \dots & \dots & \dots & \dots \\ \int_{m_{K-1}}^{m_K} E_i dt & \int_{m_{K-1}}^{m_K} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_{K-1}}^{m_K} E_j dt & \dots & \int_{m_{K-1}}^{m_K} E_j dt \end{bmatrix}$$

$(K - 1) \times (N_R + 1)$ matrix

$$\overline{x}_i = \begin{bmatrix} \alpha_i \\ k_i \\ \alpha_j \dots \\ \alpha_{N_R} \end{bmatrix}$$

$(N_R + 1) \times 1$ vector

!To solve this we need to interpolate E_i **Task** as well otherwise the integrals cannot be computed. (make sure the timelines match)

!realize that actually we have a large tensor because these matrices and vectors are regional and based on regional data. Do not be scared because we can still solve all of these matrix equations one by one and it will still give the same solution as working with tensors:)

Proof: The tensor will have the following form since for each i , $\overline{b}_i = A_i \overline{x}_i$ does not depend on other j matrices.

$$\begin{bmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & \dots & 0 \\ \dots & & & \\ 0 & 0 & \dots & A_{N_R} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_{N_R} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_{N_R} \end{bmatrix}$$

Thus as an advantage of our model's construction and setup, we can solve $\overline{b}_i = A_i \overline{x}_i$ for x_i one by one without worrying if we are satisfying tensor algebra.

After finding $P_i(t)$, $N_i(t)$, α_i , k_i , α_j : **group discussion**

! In this phase we should not force the known data that was necessary for the coefficient finding part.

! We still have to interpolate or use linear regression to use $P_i(t)$, $N_i(t)$ but at this stage we should not interpolate E_i , we interpolated E_i temporarily to solve for the coefficients,

now that the coefficients are solved we can use our differential equation model $\frac{dC_i}{dt}$ to capture the dynamic of C_i .

Now we can see how the criminal population behaves by substituting the known values $P_i(t)$, $N_i(t)$, α_i , k_i , α_j into the equation of $\frac{dC_i}{dt}$. We note that since the coefficients are found such that we match the monthly crimes committed data in the $\frac{dE_i}{dt}$ equation, our model of $\frac{dC_i}{dt}$ is highly data driven and is not approximated using empirical methods or trial and error.

One part of me thinks we can use $\frac{dE_i}{dt}$ by plugging in the coefficients to check if it can predict crimes at actual month points (monthly crimes at continuous times is weird, monthly crime rates only exist in actual months).

We can use this to:

1) See if the model predicts the future monthly crimes committed data well by simulating further in time.

2) Using the predictions of the Prophet and SARIMA models we can come up with a new Police function that achieves less crime (either using $\frac{dC_i}{dt}$ or $\frac{dE_i}{dt}$).

3) your ideas

Expressing $\frac{dC_i}{dt}$ in terms of $\frac{dE_i}{dt}$: proof-read

When we substitute $C_i = \frac{E_i}{\lambda_i}$, since λ_i is a constant, we can take the constant out of the derivative operation to express $\frac{dC_i}{dt}$ in terms of $\frac{dE_i}{dt}$. This is an important step because we have raw data on E_i .

$$\frac{dC_i}{dt} = \frac{d\frac{E_i}{\lambda_i}}{dt} = \frac{d}{dt} \left[\frac{E_i}{\lambda_i} \right] = \frac{1}{\lambda_i} \frac{d}{dt} [E_i] = \frac{1}{\lambda_i} \frac{dE_i}{dt}$$

Then we attain:

$$\frac{dC_i}{dt} = \frac{1}{\lambda_i} \frac{dE_i}{dt} = \alpha_i \frac{E_i}{\lambda_i} + \sum_{j=1, j \neq i}^{N_R} \alpha_j \frac{E_j}{\lambda_j} - k_i \frac{E_i}{\lambda_i} \left(\frac{P_i(t)}{N_i + P_i(t)} \right)$$

Since $\lambda_i > 0$, we can multiply both sides of the equation by λ_i without risking a trivial situation like $0 = 0$.

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j \frac{\lambda_i}{\lambda_j} E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

Now we assume $\lambda_i \approx \lambda_j$ for all $i \in \{1, 2, \dots, N_{Regions}\}$, $i \neq j$. Then $\frac{\lambda_i}{\lambda_j} \approx 1$. (I think $\frac{\lambda_i}{\lambda_j} < 2$ anyways)

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

Hence, we expressed $\frac{dC_i}{dt}$ in terms of E_i without relying on λ_i (we do not know λ_i and have no data for it, λ_i the average number of crimes committed in a month by a single criminal). This is a crucial step towards being able to simulate our model on Python.

Then, we find the other coefficients by “data fitting”, “trajectory fitting” or “trial and error” to make E_i go through the monthly crimes committed data that we have.

Then we return back to $\frac{dC_i}{dt}$ using $E_i = \lambda_i C_i$ and simulate it and model the behavior of the criminal population by being data driven and capturing the real life situation of UK crimes.

Coefficients: (Intuitive) (write)

α_i : Local contagion of criminal activity.

β_j : Effects of region j on region i .

k_i : Criminal catching efficiency?

λ_i : the average number of crimes committed in a month by a single criminal.

Note:

When setting up your NLLS optimizer, put strict boundaries on α_i between 0.001 and 0.5 (from now on we use λ to denote average crimes committed by a criminal in one month in any region)

Story or Intuition behind modeling $\frac{dC_i}{dt}$: “mother model” (write)

$$\frac{dC_i}{dt} = \alpha_i C_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j C_j - k_i C_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

We modeled the number of criminals as

The resulting intuition behind “the child model” $\frac{dE_i}{dt}$: (write)

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

We actually did not model the crimes committed in a month but, by the substitution $E_i = \lambda_i C_i$, we obtained this model above. Our $\frac{dC_i}{dt}$ criminal population model gave birth to the $\frac{dE_i}{dt}$ monthly crimes committed model

The monthly crimes committed increase by its present amount and the neighboring present amounts of monthly crimes committed. But it decreases by policing with diminishing returns.

Explanation for $E_i = \lambda_i C_i$: (write)

The subscript i denotes the i -th region.

$X_i(t)$ is the number of criminals at a point t in time: it is a stochastic process. $W_{k,i}(t)$ is the number of crimes the k -th criminal committed in that month, which is also a stochastic process. Then if we sum up the individual number of crimes each criminal commits in a month, it equals the total number of crimes committed in a month $Y_i(t)$.

The sum of stochastic processes is also a stochastic process, so $Y_i(t)$ is also a stochastic process.

$$\sum_{k=1}^{X_i(t)} W_{k,i}(t) = Y_i(t)$$

Let's take the expectation, $\mathbb{E}$, of both sides:

$$\mathbb{E} \left[\sum_{k=1}^{X_i(t)} W_{k,i}(t) \right] = \mathbb{E}[Y_i(t)]$$

We now use the so-called tower property:

$$\mathbb{E} \left[\mathbb{E} \left[\sum_{k=1}^{X_i(t)} W_{k,i}(t) \mid X_i(t) = n \right] \right] = \mathbb{E}[Y_i(t)]$$

By linearity of expectation:

$$\mathbb{E} \left[\sum_{k=1}^{X_i(t)} \mathbb{E}[W_{k,i}(t) \mid X_i(t) = n] \right] = \mathbb{E}[Y_i(t)] \text{ (needs work)}$$

Let's denote $\mathbb{E}[W_{k,i}(t) \mid X_i(t) = n] = \lambda_{k,i}$

$$\mathbb{E}[\sum_{k=1}^{X_i(t)} \lambda_{k,i}] = \mathbb{E}[Y_i(t)]$$

Assuming $\lambda_{k,i} \approx \lambda_{m,i}$ for all k and m not equal so $\lambda_i = \lambda_{k,i}$: (needs work)

$$\sum_{k=1}^{X_i(t)} \lambda_{k,i} = \lambda_{1,i} + \lambda_{2,i} + \dots + \lambda_{X_i(t),i} = \lambda_i + \lambda_i + \dots + \lambda_i = X_i(t) \lambda_i$$

$$\mathbb{E}[\sum_{k=1}^{X_i(t)} \lambda_{k,i}] = \mathbb{E}[X_i(t) \lambda_i] = \mathbb{E}[Y_i(t)]$$

$$\lambda_i \mathbb{E}[X_i(t)] = \mathbb{E}[Y_i(t)]$$

$$\lambda_i C_i = E_i$$

When we know $Y_i(t) = N_{C,i}$ by the data points, then since expectation of a constant equals the constant:

$$E_i = \mathbb{E}[Y_i(t)] = \mathbb{E}[N_{C,i}] = N_{C,i}$$

Expectation equals actual outcome data of $Y_i(t) = N_{C,i}$

Hence, the equation below is justified and

$$\lambda_i C_i = E_i$$

Proof of Least Squares Solution Formula: (write)

$$\bar{b}_i = A_i^T \bar{x}_i$$

Hence the equation below:

$$\bar{x}_i = (A_i^T A_i)^{-1} A_i^T \bar{b}_i$$